Budget

Amazon Research Awards, Fall 2026 — Amazon Security

Measuring Feedback-Driven Improvement in Lightweight Code-Security Verifiers for Spec-Driven Code Generation

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Cash funding needed: $79,350 USD
AWS Promotional Credits needed: up to $40,000 USD (conditional, stretch tier)

Funds Needed

We request $79,350 in cash funding to support the full 12-month execution of this project, and, conditional on award, up to $40,000 in AWS Promotional Credits as a stretch tier. Required scale evidence for this proposal is local-capacity only (verifier latency/throughput and loop-iteration cost on local GPUs); the AWS credits validate the same pipeline at larger multi-instance scale and gate no core deliverable. All requested funds are applied to direct project expenses; no institutional overhead is requested.

A. Cash Funding ($79,350)

- **Graduate Student, hourly ($34,350)**: 1,500 hours at $20/hour ($30,000 base wages), the rate this lab already uses for graduate research assistants on an active sponsored project, plus Kent's published graduate-assistant fringe rate of 14.5% ($4,350), for a fully-loaded total of $34,350. This individual leads the oracle pipeline implementation, verifier fine-tuning, and the feedback-loop workload described in the Methods section.
- **Undergraduate Student, hourly ($12,000)**: 800 hours at $15/hour, a standard Kent State undergraduate research-assistant rate (pending institutional confirmation). Covers corpus curation, harness maintenance, and evaluation support under graduate-student and PI supervision.
- **PI effort ($20,000)**: Summer-rate-equivalent continuous effort over the 12-month project, increased relative to a single-student budget to reflect direct supervision of a two-tier (graduate plus undergraduate) student team; covers system design, oracle-protocol specification, and dissemination.
- **Domestic Travel ($5,000)**: Supports travel to two conferences (e.g., IEEE S&P, USENIX Security, or ISSTA) to present interim and final (M12) results and engage with the research community.
- **Equipment / storage / oracle infra ($4,000)**: Sandboxes, CPU fuzz instances, and CodeQL/Semgrep compute for the oracle battery; institutional (non-AWS) compute that complements the AWS credits request.
- **External inference ($3,000)**: Frontier LLM judge baseline (API) used for the cost-quality Pareto comparison in the Evaluation section; estimate, revisited at the M1 pilot cost measurement.
- **Contingency ($1,000)**: Unallocated buffer for unforeseen direct costs.

The cash request allocates direct expenses for personnel, travel, infrastructure, and external inference; it does not establish fully loaded personnel coverage beyond the graduate-assistant fringe included above. Single-institution (Kent) solo budget: graduate student support (the 1,500h are shared across appointments, not one person's schedule), one undergraduate student, and PI effort; no Co-PI, no sub-award. Graduate wages include Kent's published 14.5% graduate-assistant fringe rate; undergraduate wages, tuition, and fully-loaded PI treatment remain pending institutional confirmation and are excluded from the cash total above. No institutional commitment to cover these remaining charges is documented; the proposed 1,500 graduate and 800 undergraduate hours therefore remain wage-based estimates apart from the included graduate fringe. PI hours are not documented. Mandatory charges and reserved compute capacity must be reconciled before freezing core scope; full personnel coverage is unconfirmed.

The core experiment comprises two policies $\times$ three paired seeds $\times$ ten rounds (60 adapter updates), one shared initial-adapter fit per seed (three total), and 66 policy/seed checkpoints including $t = 0$, plus final evaluation. Local GPU hours are uncommitted and no measured GPU/CPU-hour estimate is available. M1 must cost generation, initial fitting, all updates/checkpoint scoring, and the oracle battery against reserved hardware/hours; conditional AWS credits are not a core compute commitment. Within the personnel totals, reserve 300 graduate hours and 200 undergraduate hours for harness construction and independent validation (wages \$6,000 plus \$3,000); PI adjudication uses PI effort, with actual time recorded. The \$9,000 harness/replay allocation (\$6,000 graduate and \$3,000 undergraduate wages) is included in the personnel totals, not additional funds; 1,200 graduate and 600 undergraduate hours remain for curation, model updates, and evaluation. These allocations are estimates rather than measured validation costs.

B. AWS Promotional Credits (up to $40,000, conditional stretch tier)

To validate the verifier and feedback loop at AWS production scale beyond the local-GPU pilot, we request AWS credits allocated as follows:

- **Amazon SageMaker ($20,000)**: Multi-arm LoRA fine-tuning and evaluation of the verifier at scale, covering the main feedback-loop and ablation-arm training workload.
- **Amazon EC2, compute-optimized ($7,000)**: CPU-bound fuzz/eval farm (Atheris / Jazzer.js harnesses) for the oracle battery at scale.
- **Amazon EC2, g-series (GPU) ($5,000)**: GPU-bound inference and evaluation of the verifier at scale, complementing the SageMaker training allocation.
- **Amazon S3 + data transfer ($4,000)**: Corpus, traces, and artifact storage, plus inter-service data transfer.
- **AWS CloudWatch ($2,000)**: Pipeline monitoring and run-level logging across the scaled oracle and training jobs.
- **Buffer ($2,000)**: Reserved for instance overage and S3 egress as experimental scale is tuned.

AWS credits are for validating the verifier and feedback loop under realistic production-scale conditions; they are stretch-only, since the core plan excludes dependence on AWS credits, although local capacity is still uncommitted and must be established at M1. Released safe-fixture subset results are reproducible from public artifacts; full-corpus results additionally require upstream retrieval under recorded licenses. AWS Public Dataset usage: N/A — the corpus is self-generated (public-repository task instances plus CVE-sourced and seeded-bug instances built by this project), so there is no public-dataset dependency. The allocations above are estimated shares, not unit prices; M1 fixes core/stretch experiment counts and per-product unit-cost assumptions, revisited if realized costs diverge, consistent with the Proposal document.